

Администрирование локальных сетей

Лабораторная работа №10

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Цели и задачи работы

Освоить настройку прав доступа пользователей к ресурсам сети и изучить применение списков управления доступом (ACL) в Cisco Packet Tracer.

Ход выполнения работы

Подключение ноутбука администратора

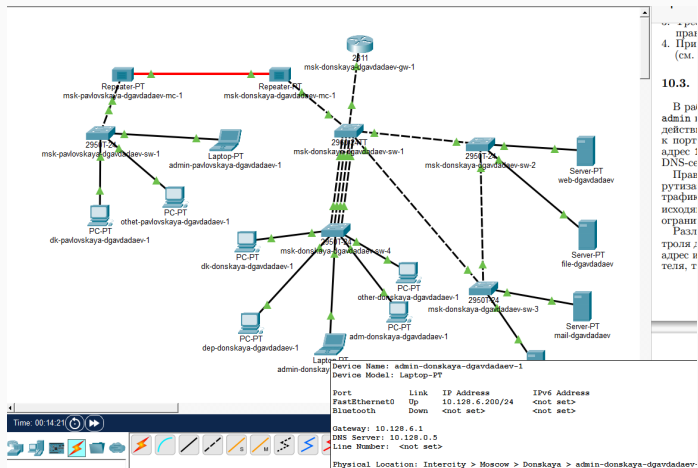


Рис. 1: Подключение и настройка ноутбука администратора

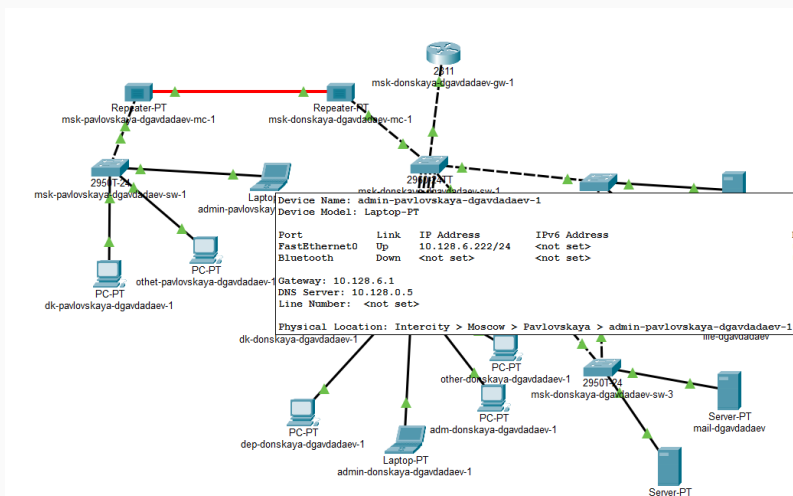


Рис. 2: Топология сети

```
Password:
msk-donskaya-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-dgavdadaev-gw-1(config)#
msk-donskaya-dgavdadaev-gw-1(config)#ip access-list extended servers-out
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark web
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp any host
10.128.0.2 eq 80
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#exit
msk-donskaya-dgavdadaev-gw-1(config)#int f0/0.3
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip access-group servers-out out
msk-donskaya-dgavdadaev-gw-1(config-subif)#
```

Рис. 3: Создание ACL для web-доступа

```
Password:
msk-donskaya-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-dgavdadaev-gw-1(config)#
msk-donskaya-dgavdadaev-gw-1(config)#ip access-list extended servers-out
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark web
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp any host
10.128.0.2 eq 80
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#exit
msk-donskaya-dgavdadaev-gw-1(config)#int f0/0.3
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip access-group servers-out out
msk-donskaya-dgavdadaev-gw-1(config-subif)#
```

Рис. 4: Применение ACL к интерфейсу

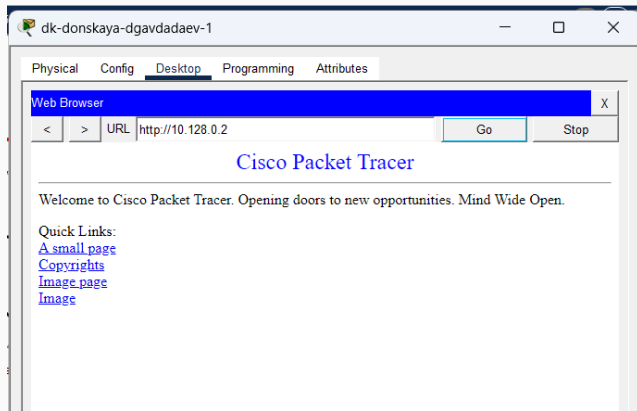


Рис. 5: Проверка доступа через браузер

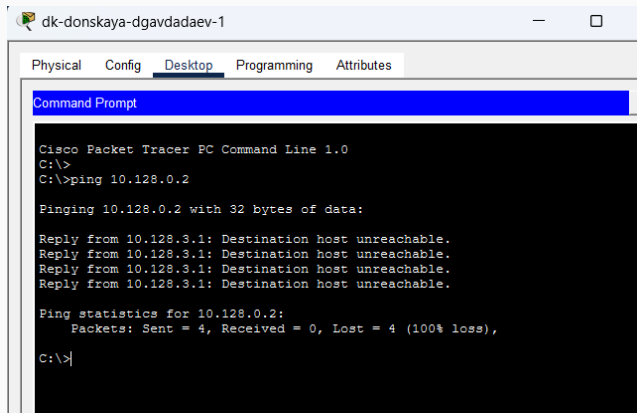
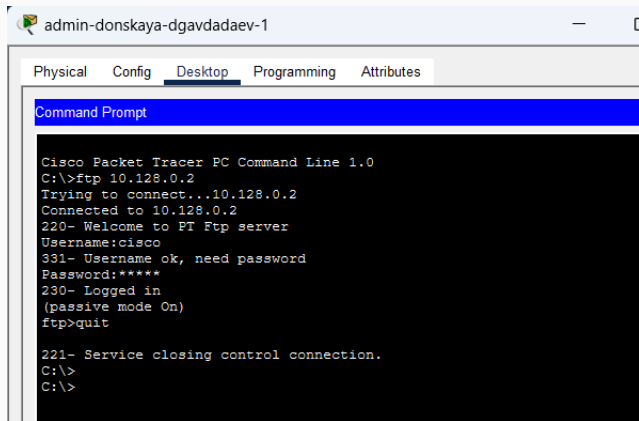


Рис. 6: Проверка через ping

```
msk-donskaya-dgavdadaev-gw-1(config)#  
msk-donskaya-dgavdadaev-gw-1(config)#ip access-list extended servers-out  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp host 10.128.6.200  
host 10.128.0.2 range 20 ftp  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp host 10.128.6.222  
host 10.128.0.2 range 20 ftp  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp host 10.128.6.200  
host 10.128.0.2 eq telnet  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp host 10.128.6.222  
host 10.128.0.2 eq telnet  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#
```

Рис. 7: Добавление правил для администратора

Проверка FTP-доступа администратора



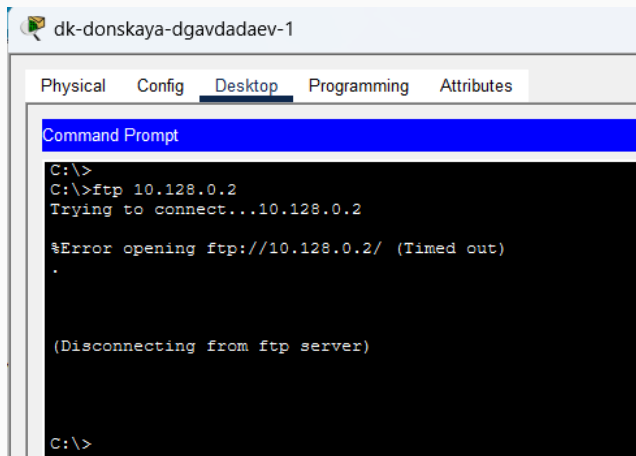
The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'admin-donskaya-dgavdadaev-1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The command prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
Connected to 10.128.0.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>quit

221- Service closing control connection.
C:\>
C:\>
```

Рис. 8: FTP-доступ администратора

Проверка ограничения для других пользователей



The screenshot shows a MikroTik WinBox window titled "dk-donskaya-dgavdadaev-1". The "Desktop" tab is selected. A Command Prompt window is open, displaying the following text:

```
C:\>  
C:\>ftp 10.128.0.2  
Trying to connect...10.128.0.2  
  
%Error opening ftp://10.128.0.2/ (Timed out)  
.  
  
(Disconnecting from ftp server)  
  
C:\>
```

Рис. 9: Отказ в доступе для других пользователей

Настройка доступа к файловому серверу

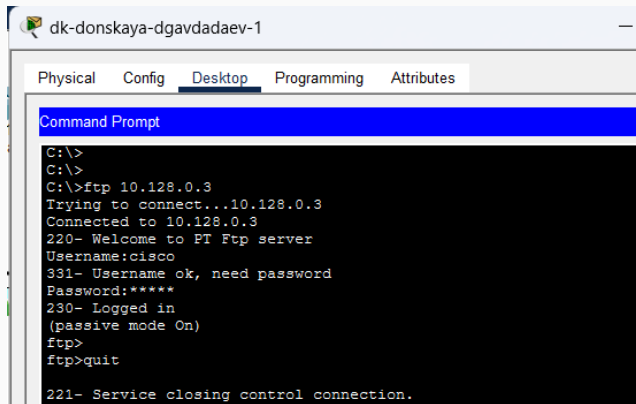
```
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark file  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp 10.128.0.0  
0.0.255.255 host 10.128.0.3 eq 445  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp any host  
10.128.0.3 range 20 ftp  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#
```

Copy

Paste

Рис. 10: Настройка доступа к файловому серверу

Проверка доступа к файловому серверу



```
Command Prompt
C:\>
C:\>
C:\>ftp 10.128.0.3
Trying to connect...10.128.0.3
Connected to 10.128.0.3
220- Welcome to FT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
ftp>quit
221- Service closing control connection.
```

Рис. 11: Проверка FTP-доступа к файловому серверу

```
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark mail  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp any host  
10.128.0.4 eq smtp  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit tcp any host  
10.128.0.4 eq pop3  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark dns  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit udp 10.128.0.0  
0.0.255.255 host 10.128.0.5 eq 53  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#1 permit icmp any any  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#
```

Рис. 12: Настройка доступа к почтовому серверу и DNS


```
msk-donskaya-dgavdadaev-gw-1#sn ac
msk-donskaya-dgavdadaev-gw-1#sh access-lists
Extended IP access list servers-out
  1 permit icmp any any
  10 permit tcp any host 10.128.0.2 eq www (6 match(es))
  20 permit tcp host 10.128.6.200 host 10.128.0.2 range 20 ftp (10
match(es))
  30 permit tcp host 10.128.6.222 host 10.128.0.2 range 20 ftp
  40 permit tcp host 10.128.6.200 host 10.128.0.2 eq telnet
  50 permit tcp host 10.128.6.222 host 10.128.0.2 eq telnet
  60 permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445
  70 permit tcp any host 10.128.0.3 range 20 ftp (11 match(es))
  80 permit tcp any host 10.128.0.4 eq smtp
  90 permit tcp any host 10.128.0.4 eq pop3
  100 permit udp 10.128.0.0 0.0.255.255 host 10.128.0.5 eq domain
Extended IP access list other-in
  10 permit ip host 10.128.6.200 any
  20 permit ip host 10.128.6.222 any
Extended IP access list management-out
  10 permit ip host 10.128.6.200 10.128.1.0 0.0.0.255
  20 permit ip host 10.128.6.222 10.128.1.0 0.0.0.255
msk-donskaya-dgavdadaev-gw-1#
```

Рис. 13: Проверка списка ACL

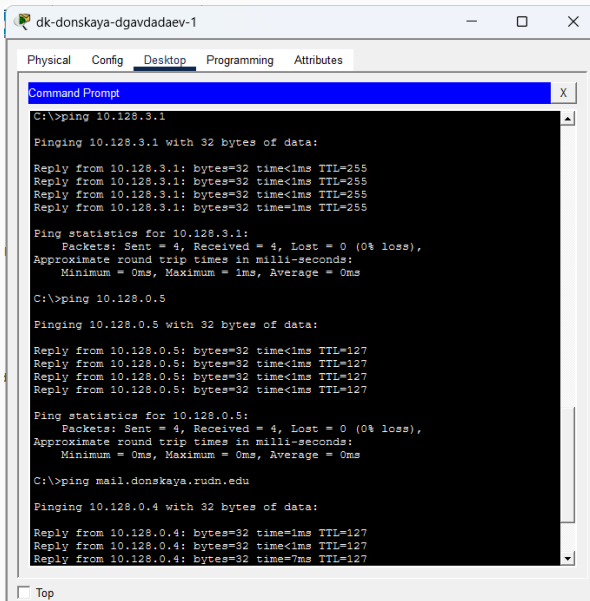
```
msk-donskaya-dgavdadaev-gw-1(config)#  
msk-donskaya-dgavdadaev-gw-1(config)#ip access-list extended other-in  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark admin  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit ip host 10.128.6.200  
any  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit ip host 10.128.6.222  
any  
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#exit  
msk-donskaya-dgavdadaev-gw-1(config)#int f0/0.104  
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip access-group other-in in  
msk-donskaya-dgavdadaev-gw-1(config-subif)#  
msk-donskaya-dgavdadaev-gw-1(config-subif)#
```

Рис. 14: Настройка ACL для сети Other

```
msk-donskaya-dgavdadaev-gw-1(config)#
msk-donskaya-dgavdadaev-gw-1(config)#ip access-list extended management-out
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark admin
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit ip host 10.128.6.200
10.128.1.0 0.0.0.255
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit ip host 10.128.6.222
10.128.1.0 0.0.0.255
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#exit
msk-donskaya-dgavdadaev-gw-1(config)#int f0/0.2
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip access-group management-out
out
msk-donskaya-dgavdadaev-gw-1(config-subif)#exit
msk-donskaya-dgavdadaev-gw-1(config)#exit
```

Рис. 15: Настройка доступа к сети управления

Проверка работы сетевых сервисов



The screenshot shows a Windows Command Prompt window titled "dk-donskaya-dgavdadaev-1". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" selected. The Command Prompt displays the following output:

```
C:\>ping 10.128.3.1

Pinging 10.128.3.1 with 32 bytes of data:

Reply from 10.128.3.1: bytes=32 time<1ms TTL=255
Reply from 10.128.3.1: bytes=32 time<1ms TTL=255
Reply from 10.128.3.1: bytes=32 time<1ms TTL=255
Reply from 10.128.3.1: bytes=32 time=1ms TTL=255

Ping statistics for 10.128.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.0.5

Pinging 10.128.0.5 with 32 bytes of data:

Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127
Reply from 10.128.0.5: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping mail.donskaya.rudn.edu

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=7ms TTL=127
```

At the bottom left of the window, there is a checkbox labeled "Top" which is currently unchecked.

Проверка web-доступа по имени

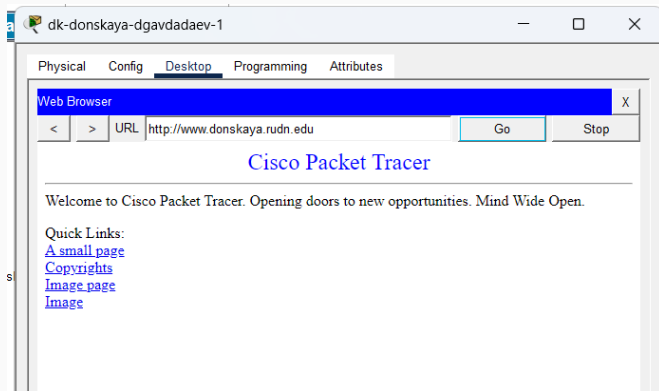


Рис. 17: Проверка web-доступа по имени

Проверка работы почтового сервиса

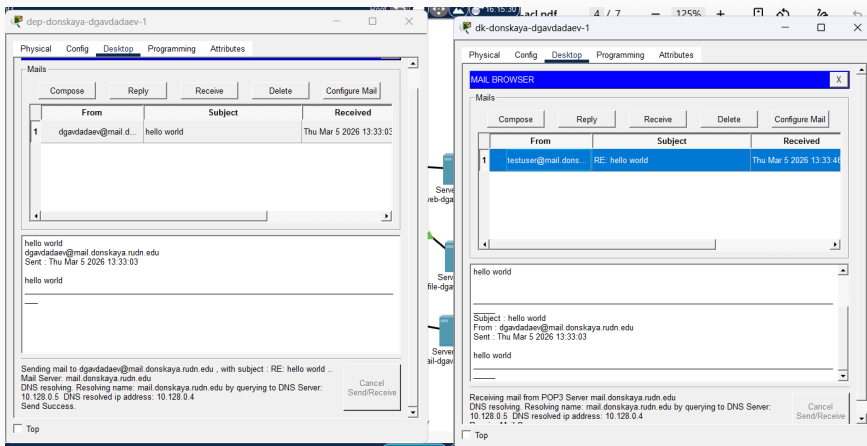
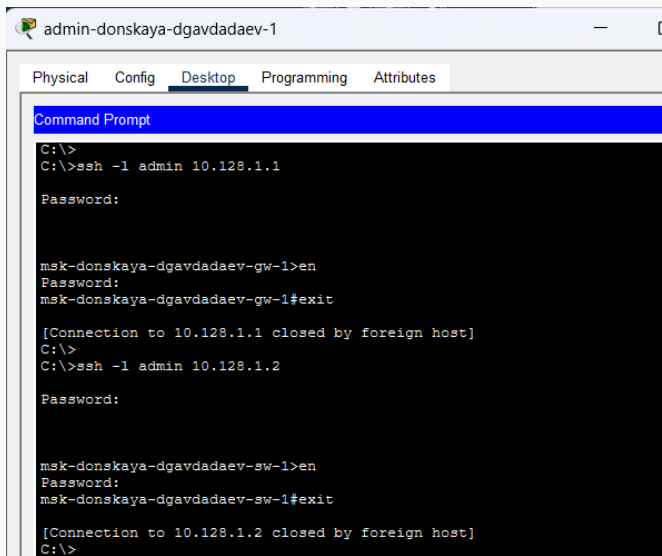


Рис. 18: Проверка почтового сервиса

Проверка SSH-доступа администратора



The screenshot shows a Windows desktop environment. At the top, there is a taskbar with the text 'admin-donskaya-dgavdadaev-1'. Below the taskbar, there are several tabs: 'Physical', 'Config', 'Desktop' (which is selected), 'Programming', and 'Attributes'. The main content area displays a 'Command Prompt' window with a blue title bar. The command prompt shows the following sequence of commands and outputs:

```
C:\>  
C:\>ssh -l admin 10.128.1.1  
  
Password:  
  
msk-donskaya-dgavdadaev-gw-1>en  
Password:  
msk-donskaya-dgavdadaev-gw-1#exit  
  
[Connection to 10.128.1.1 closed by foreign host]  
C:\>  
C:\>ssh -l admin 10.128.1.2  
  
Password:  
  
msk-donskaya-dgavdadaev-sw-1>en  
Password:  
msk-donskaya-dgavdadaev-sw-1#exit  
  
[Connection to 10.128.1.2 closed by foreign host]  
C:\>
```

Выводы

В ходе выполнения лабораторной работы:

- настроены списки управления доступом ACL
- организован доступ к web, file, mail и DNS-серверам
- реализованы ограничения для пользовательских сегментов сети
- настроен отдельный доступ администратора к сетевым ресурсам
- проверена работа HTTP, FTP, SMTP, POP3, DNS, ICMP и SSH